THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A chord of a circle of radius 7 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 4.47 cm² **(b)** 25.67 cm² **(c)** 46.86 cm² **(d)** 21.19 cm²

2. A brooch is a circular piece of silver wire of diameter 21 mm. The wire is also used for 4 equal diameters that divide the brooch into 8 sectors. The total length of wire required (in mm) is: 1

(a) 108 mm **(b)** 66 mm **(c)** 346.5 mm **(d)** 150 mm

3. A horse is tied at the midpoint of one side of a large square field with a rope 35 m long. The area of the field it can graze is: 1

(a) 1925 m² **(b)** 110 m² **(c)** 3850 m² **(d)** 55 m²

4. A sector of a circle of radius 42 cm has central angle 45°. The perimeter of the sector is: 1

(a) 117 cm **(b)** 33 cm **(c)** 159 cm **(d)** 75 cm

5. An arc of length 22 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 150° **(b)** 60° **(c)** 30° **(d)** 15°

6. A square is inscribed in a circle of radius 35 cm. The area of the circle outside the square is: 1

(a) 3850 cm² **(b)** 2450 cm² **(c)** 2625 cm² **(d)** 1400 cm²

7. A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 35 m. The increase in the area it can graze is: 1

(a) 3696 m² **(b)** 962.5 m² **(c)** 616 m² **(d)** 924 m²

8. A chord of a circle of radius 10.5 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 163.18 cm² **(b)** 115.50 cm² **(c)** 47.68 cm² **(d)** 67.82 cm²

9. **Assertion (A):** The area of a sector of a circle of radius 7 cm and angle 90° is $38.5\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The area of a sector of angle θ is $\dfrac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The length of the arc of a sector of radius 21 cm and angle 60° is 22 cm (use $\pi = \frac{22}{7}$). 1

Reason (R): The length of the arc of a sector of angle θ is $\frac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An umbrella has 14 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 49 cm. Find the area between two consecutive ribs. 2

12. A wheel has radius 35 cm. Find the number of complete revolutions it makes to cover 110 m. 2

13. A circular park of radius 35 m is surrounded by a path 3.5 m wide. Find the area of the path. 2

14. An umbrella has 18 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 63 cm. Find the area between two consecutive ribs. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sector of a circle of radius 42 cm has central angle 50°. Find the area of the sector. 3

16. A quadrant of a circle of radius 63 cm is cut from a square of side 63 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

17. The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 5 minutes. 3

OR

A sector of a circle of radius 42 cm has central angle 80°. Find the area of the sector. 3

18. A quadrant of a circle of radius 70 cm is cut from a square of side 70 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

OR

The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 10 minutes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A lawn sprinkler is fixed at a point and waters a sector-shaped region of radius 21 m with a sector angle of 60°. (Take $\pi = \frac{22}{7}$.) 5

(i) Find the area of the lawn watered.

(ii) Find the length of the curved boundary (arc).

(iii) Find the perimeter of the watered region.

(iv) If the sprinkler is adjusted so that the angle becomes 120°, find the new area and compare it with the old one.

20. A chord of a circle of radius 42 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 72° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER — **Total: 40 Marks**